



Global Migration of Scholars: Trends and Patterns revealed by Bibliometric Data

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Use of bibliometric data for demographic research and study of scholarly migration at the MPIDR:

- Zhao, X., Aref, S., Zagheni, E., & Stecklov, G. (2022). Return migration of German-affiliated researchers: Analyzing departure and return by gender, cohort, and discipline using Scopus bibliometric data 1996-2020. *Scientometrics*
- Kashyap, R., Rinderknecht, R. G., Akbaritabar, A., Alburez-Gutierrez, D., Gil-Clavel, S., Grow, A., Kim, J., Leasure, D. R., Lohmann, S., Negraia, D. V., Perrotta, D., Rampazzo, F., Tsai, C.-J., Verhagen, M. D., Zagheni, E., & Zhao, X. (2022). *Digital and Computational Demography*. SocArXiv.
<https://doi.org/10.31235/osf.io/7bvpt>
- Zhao, X., Aref, S., Zagheni, E., & Stecklov, G. (2021a). International Migration in Academia and Citation Performance: An Analysis of German-Affiliated Researchers by Gender and Discipline Using Scopus Publications 1996-2020. ArXiv:2104.12380 [Cs]. <http://arxiv.org/abs/2104.12380>
- Subbotin, A., & Aref, S. (2021). Brain drain and brain gain in Russia: Analyzing international migration of researchers by discipline using Scopus bibliometric data 1996–2020. *Scientometrics*. <https://doi.org/10.1007/s11192-021-04091-x>
- Miranda-González, A., Aref, S., Theile, T., & Zagheni, E. (2020). Scholarly migration within Mexico: Analyzing internal migration among researchers using Scopus longitudinal bibliometric data. *EPJ Data Science*, 9(1), 34. <https://doi.org/10.1140/epjds/s13688-020-00252-9>
- Aref, S., Zagheni, E., & West, J. (2019). The Demography of the Peripatetic Researcher: Evidence on Highly Mobile Scholars from the Web of Science. In I. Weber, K. M. Darwish, C. Wagner, E. Zagheni, L. Nelson, S. Aref, & F. Flöck (Eds.), *Social Informatics* (pp. 50–65). Springer International Publishing.
https://doi.org/10.1007/978-3-030-34971-4_4
- Alburez-Gutierrez, D., Zagheni, E., Aref, S., Gil-Clavel, S., Grow, A., & Negraia, D. V. (2019). Demography in the digital era: New data sources for population research.
- Abel, G. J., Muttarak, R., Bordone, V., & Zagheni, E. (2019). Bowling Together: Scientific Collaboration Networks of Demographers at European Population Conferences. *European Journal of Population*, 35(3), 543–562. <https://doi.org/10.1007/s10680-018-9493-1>

HUMAN BIOLOGY

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VOL. 6



No. 4

FERTILITY OF THE SOCIAL CLASSES IN THE NATIVE WHITE POPULATION OF COLUMBUS AND SYRACUSE¹

BY FRANK W. NOTESTEIN AND CLYDE V. KISER
Milbank Memorial Fund

ONE of the neglected aspects of the much discussed decline in the birth rate is the problem of the incidence of the decline in constituent classes of the population. We know that the birth rate has dropped more rapidly in the foreign than in the native population. But within the native group we know little enough of the nature of the changes that have taken place. Has the decline fallen with full force on the established families which have remained unbroken? Within this group are the birth rates declining with greater rapidity in some classes than in others, and if so, are these differences tending to strengthen or weaken the well-established inverse relation between social status and fertility?

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ECONOMIC, FINANCIAL AND TRANSIT
DEPARTMENT



The Future Population of Europe and the Soviet Union Population Projections 1940-1970

BY

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PRINCETON UNIVERSITY

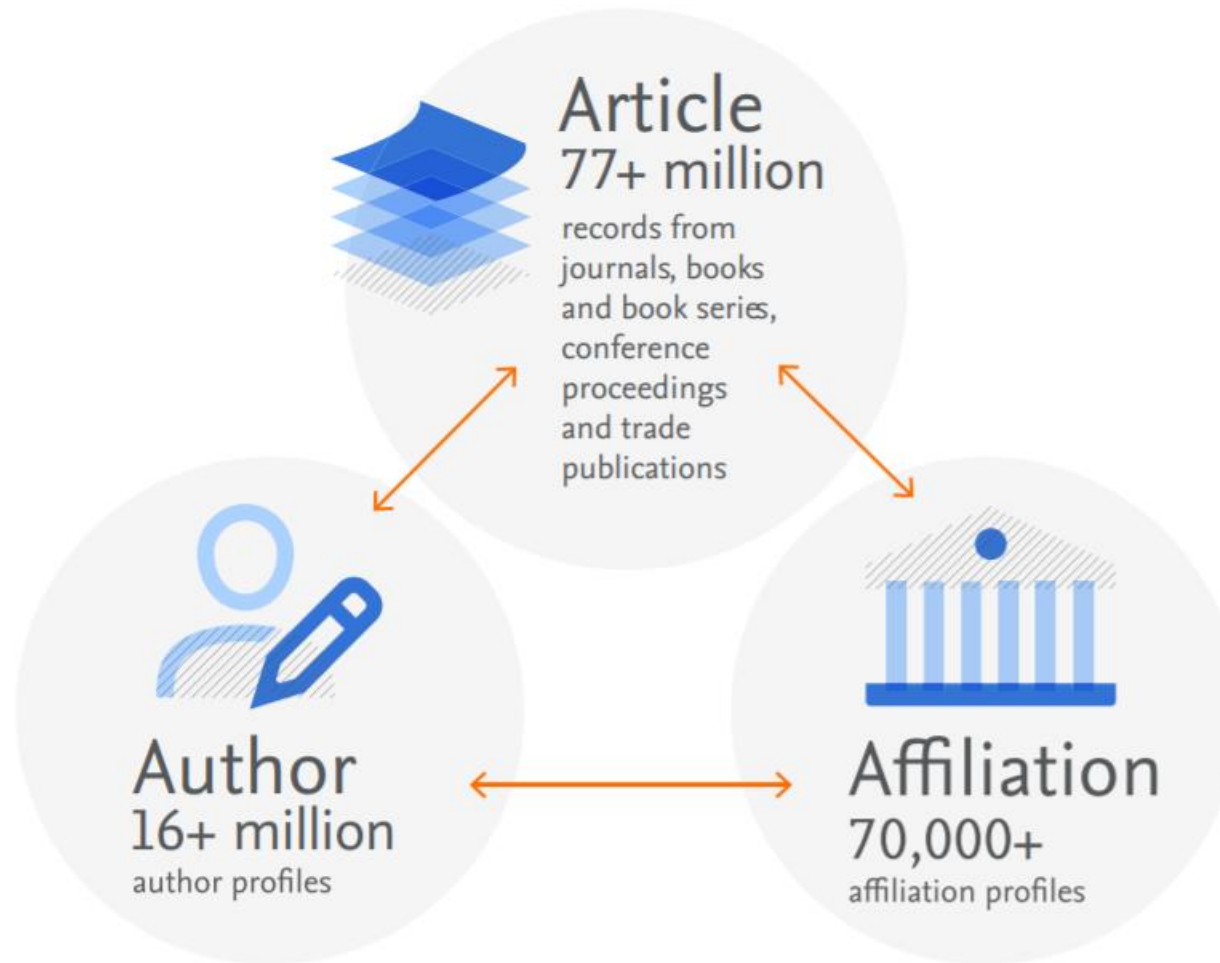
LEAGUE OF NATIONS, GENEVA, 1944

A SIMPLE IDEA

Changes in institutional affiliations can be used to infer changes in residence over time for individual scholars and for populations

Publication Title	Author Full Name	Affiliation	Publication Year
Paper 1 title	John Doe	Affiliation 1, Country 1	2020
Paper 1 title	John Doe	Affiliation 2, Country 2	2020
Paper 1 title	Jane Doe	Affiliation 2, Country 2	2020

A SCALABLE IDEA: THE SCOPUS DATABASE



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NEW INSIGHTS AT THE MICRO AND MACRO LEVELS



OUTLINE

- **BACKGROUND ON THE DATA**
- **MIGRATION, AND DEVELOPMENT**

OUTLINE

- **BACKGROUND ON THE DATA**
- **MIGRATION, AND DEVELOPMENT**

AN EXAMPLE OF THE TYPE OF DATA



authorsid	authorswithaffiliations	year	country
23494432900	Zagheni, E., Laboratory of Digital and Computational Demography, Max Planck Institute for Demographic Research, Rostock, Germany	2020	DEU
23494432900	Zagheni, E., Laboratory of Digital and Computational Demography, Max Planck Institute for Demographic Research, Rostock, Germany	2020	DEU
23494432900	Zagheni, E., Max Planck Institute for Demographic Research Rostock, Germany	2020	DEU
23494432900	Zagheni, E., Max Planck Institute for Demographic Research, Germany	2020	DEU
23494432900	Zagheni, E., Max Planck Institute for Demographic Research, Germany	2020	DEU
23494432900	Zagheni, E., Max Planck Institute for Demographic Research, Rostock, Germany	2020	DEU
23494432900	Zagheni, E., Max Planck Institute for Demographic Research, Rostock, Germany	2019	DEU
23494432900	Zagheni, E., Max Planck Institute for Demographic Research, Konrad-Zuse-Straße 1, Rostock, 18057, Germany	2019	DEU
23494432900	Zagheni, E.	2019	
23494432900	Zagheni, E., Department of Sociology, University of Washington, Seattle, United States, Max Planck Institute for Demographic Research, Rostock, Germany	2019	DEU
23494432900	Zagheni, E., Max Planck Institute for Demographic Research, Germany	2019	DEU
23494432900	Zagheni, E., Max Planck Institute for Demographic Research, Rostock, Germany	2019	DEU
23494432900	Zagheni, E., Max Planck Institute for Demographic Research, Germany	2019	DEU
23494432900	Zagheni, E., Max Planck Institute for Demographic Research, Konrad-Zuse-Str. 1, Rostock, 18057, Germany	2019	DEU
23494432900	Zagheni, E., Qatar Computing Research Institute, Doha, Qatar	2019	QAT
23494432900	Zagheni, E., Max Planck Institute for Demographic Research	2019	
23494432900	Zagheni, E., Max Planck Institute for Demographic Research, Rostock, Germany	2018	DEU
23494432900	Zagheni, E., Max Planck Institute for Demographic Research, Rostock, Germany	2018	DEU
23494432900	Zagheni, E., University of Washington, Max Planck Institute for Demographic Research, United States	2018	USA
23494432900	Zagheni, E., University of Washington and Max Planck Institute for Demographic Research, United States	2018	USA
23494432900	Zagheni, E.	2017	
23494432900	Zagheni, E., Department of Sociology, University of Washington, Seattle, 211 Savery Hall, Box 353340, Seattle, WA 98195-3340, United States	2017	USA
23494432900	Zagheni, E., University of Washington, Seattle, United States	2017	USA
23494432900	Zagheni, E., University of Washington, United States	2017	USA
23494432900	Zagheni, E., University of Washington, Seattle, United States	2016	USA
23494432900	Zagheni, E., Department of Sociology, University of Washington at Seattle, 211 Savery Hall Box 353340, Seattle, WA 98195-3340, United States	2016	USA
23494432900	Zagheni, E., University of Washington, Seattle, WA, United States	2015	USA
23494432900	Zagheni, E., Department of Sociology, University of Washington, United States	2015	USA
23494432900	Zagheni, E., University of Washington, Seattle, United States	2015	USA
23494432900	Zagheni, E., Department of Sociology, University of Washington, Seattle, United States	2015	USA
23494432900	Zagheni, E., Department of Sociology, University of Washington, Seattle, WA, United States	2015	USA

A publication on publisher's website (left) and Scopus (right)



Leaving Home and Entering the Housing Market

William A V Clark, Clara H Mulder

First Published September 2019
<https://doi.org/10.1068/a3315>

Article information

Abstract

We use a multinomial choice model of owning a home, owning a trailer, or renting to examine the housing-market entry of young adults in the USA after they have left the parental home. We also model the choice between renting independently and sharing with roommates. We show that the likelihood of becoming an independent actor in the housing market is closely related to the size and regional location of the housing market. The young adult's resources are an important influence on housing-market entry. Parents' resources seem to be less important as a factor in housing selection. Whereas trailer ownership is more common among the less well educated, couples, and those leaving home to live in the South or in rural areas, sharing is typical for the younger nest leavers, singles, and those leaving home to live in the cities.

References

Avery, R, Goldscheider, F, Speare, A, 1992, "Feathered nest/gilded cage: Parental income and leaving home in the transition to adulthood" *Demography* 29 375-388
[Google Scholar](#) | [Crossref](#) | [Medline](#) | [ISI](#)

Börsch-Supan, A, 1986, "Household formation, housing prices, and public policy impacts" *Journal of Public Economics* 30 145-164
[Google Scholar](#) | [Crossref](#) | [ISI](#)

Buck, N, Scott, J, 1993, "She's leaving home: But why? An analysis of young people leaving the parental home" *Journal of Marriage and the Family* 55 863-874
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Leaving home and entering the housing market

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Social Science Research, 2021

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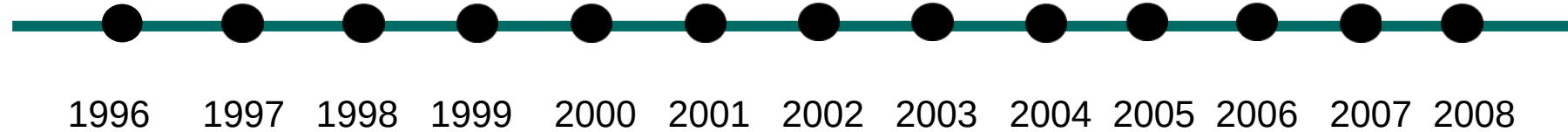
42 h-index

4,345 Citations by 2357 documents

DATA QUALITY

- **Author disambiguation in Scopus:**
98.1% of author profiles do not include publications written by someone else**;
94.4% of author profiles include all publications written by the author.
- **Organization disambiguation:**
Research Organization Registry (ROR) API
- **Subset of data with the highest quality:**
Period:1996-2020; Type of publications: Articles and Reviews;
→ 36+ Million publications for 16+ Million unique authors

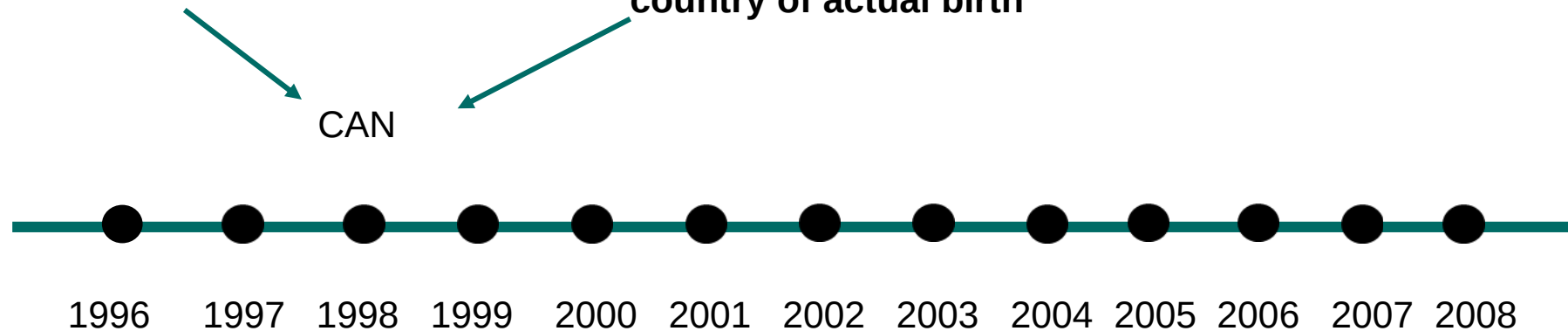
IDENTIFYING MIGRATION EVENTS: ILLUSTRATIVE EXAMPLES



IDENTIFYING MIGRATION EVENTS: ILLUSTRATIVE EXAMPLES

Maria's **first**
publications are
with a Canadian
affiliation in 1998

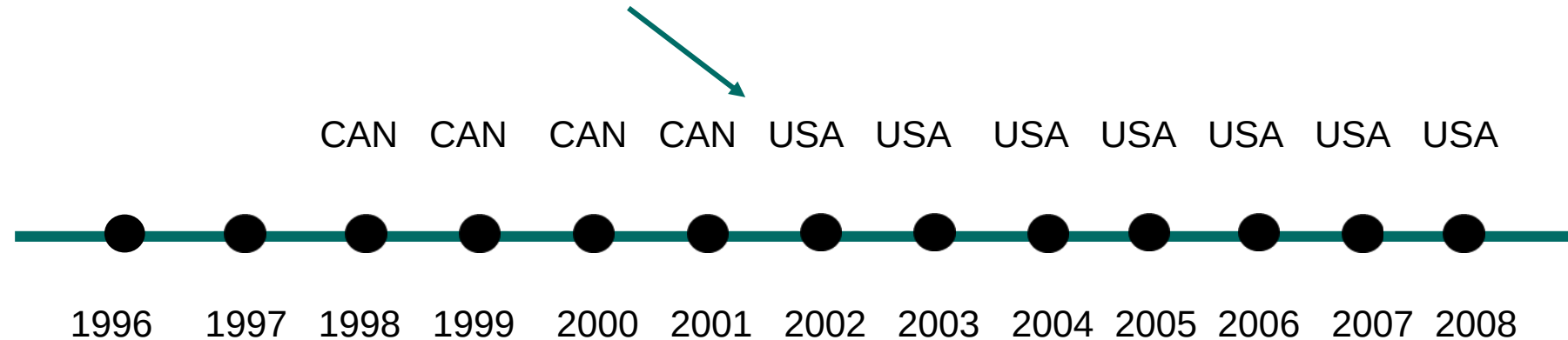
Maria's country of "**academic**" birth is
Canada, regardless of her citizenship or
country of actual birth



The country of residence
is inferred as the **modal**
country of publications
in a given year

IDENTIFYING MIGRATION EVENTS: ILLUSTRATIVE EXAMPLES

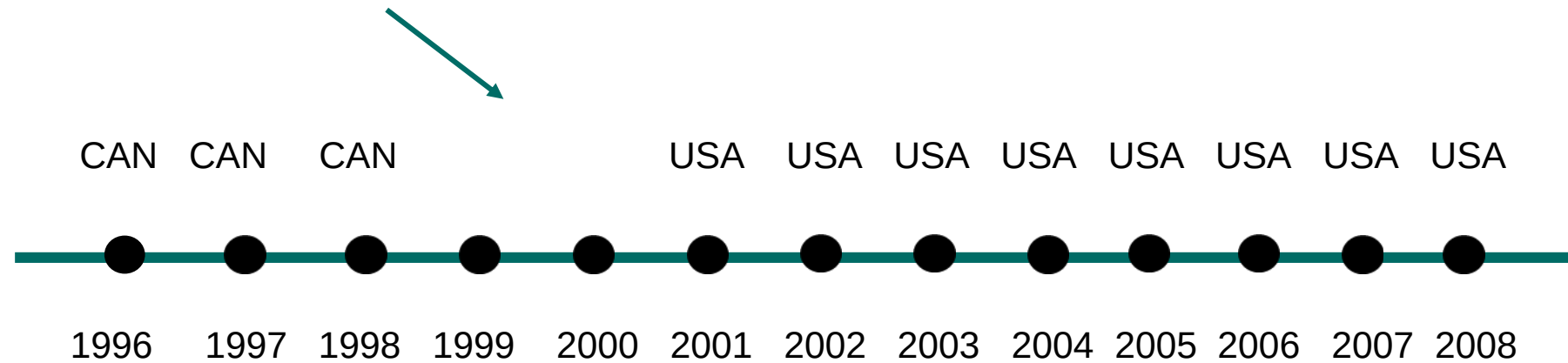
Maria changes residence
from Canada to the US
between 2001 and 2002



Maria's **modal countries**
of publications over time

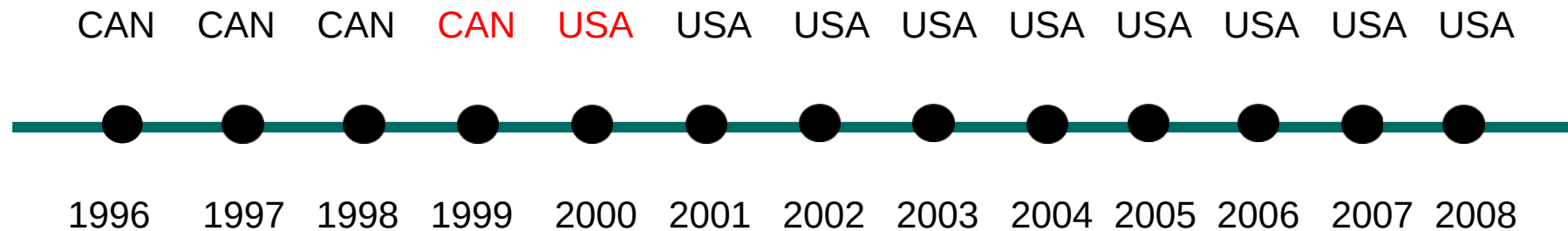
IDENTIFYING MIGRATION EVENTS: ILLUSTRATIVE EXAMPLES

Joe has no publications in our
database for 1999 and 2000



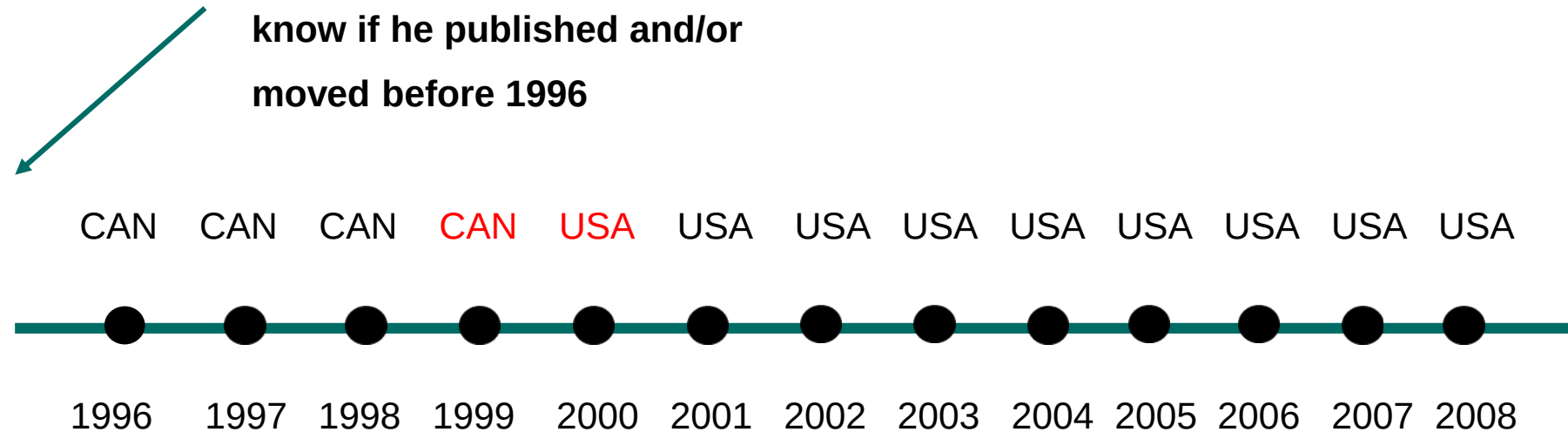
IDENTIFYING MIGRATION EVENTS: ILLUSTRATIVE EXAMPLES

We impute the country values
for up to 2 years from the
closest observation with
backward or forward filling



IDENTIFYING MIGRATION EVENTS: ILLUSTRATIVE EXAMPLES

For Joe, we may have a left-censoring issue: we do not know if he published and/or moved before 1996



OUTLINE

- **BACKGROUND ON THE DATA**
- **MIGRATION, AND DEVELOPMENT**

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- **BACKGROUND ON THE DATA**
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DOES ECONOMIC DEVELOPMENT INCREASE OR REDUCE EMIGRATION OF ACADEMIC SCIENTISTS?

- Extensive literature on migration and development for entire populations: e.g., development displaces labor from agrarian societies → in the short term migration increases, then it decreases.
- But migration processes are heterogeneous
- Theory on high-skilled migration provides some guidance (e.g., neoclassical theory, relative deprivation perspective, world systems)
- Here we focus on one specific group (active academic scientists), that is important for innovation and circulation of ideas, and for which we have prepared good “longitudinal” data.

STATISTICAL ANALYSIS

- Generalize Additive Mixed Model (GAMM) that allows to model non-linear relationships with penalized smooth functions, “s()”, of selected predictors:

$$\log \text{ emigr. counts} = \beta_0 + s(\log \text{ GDP}) + s(\text{year}) + s(\text{country}) + s(\text{country}, \log \text{ GDP}) + \log \text{ exposure} +$$

e

Smooth functions
of log(GDP) and
year

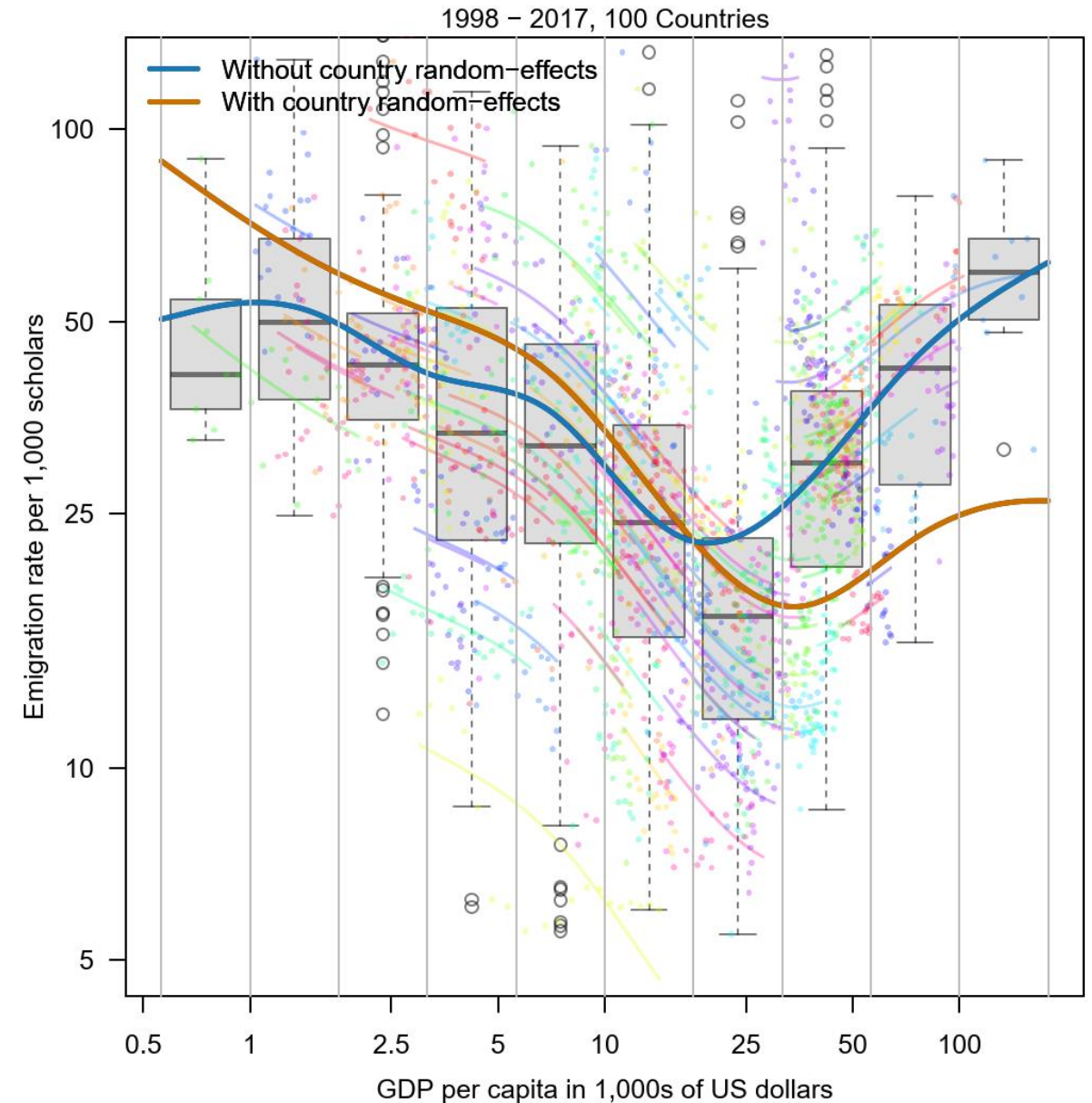
Random intercept
for countries

Random slope
for countries

DATA AND PREDICTED OUTCOMES

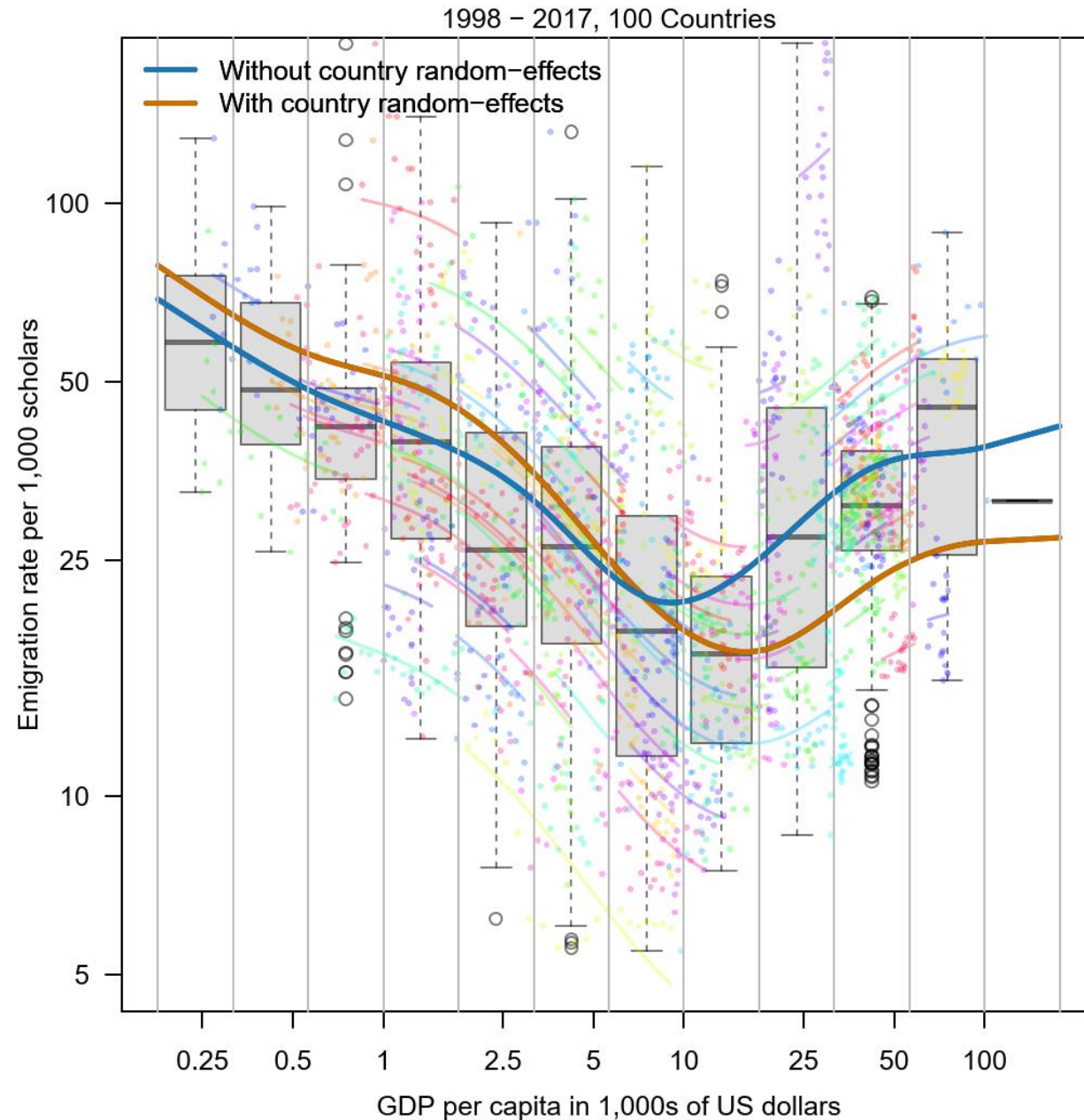
1998-2017 averaged emigration rates predicted by the model as a function of GDP per capita measured in 2017 constant international dollars at purchasing power parity.

The combined data for each year and country is plotted as a box plot, while the age- and year- specific data are plotted using colored dots. Each color represents a separate country.



DATA AND PREDICTED OUTCOMES (sensitivity analysis)

1998-2017 averaged emigration rates predicted by the model as a function of GDP per capita measured in 2010 constant international dollars



DATA AND PREDICTED OUTCOMES (sensitivity analysis)

1998-2017 averaged emigration rates predicted by the model as a function of GDP per capita measured at purchasing power parity

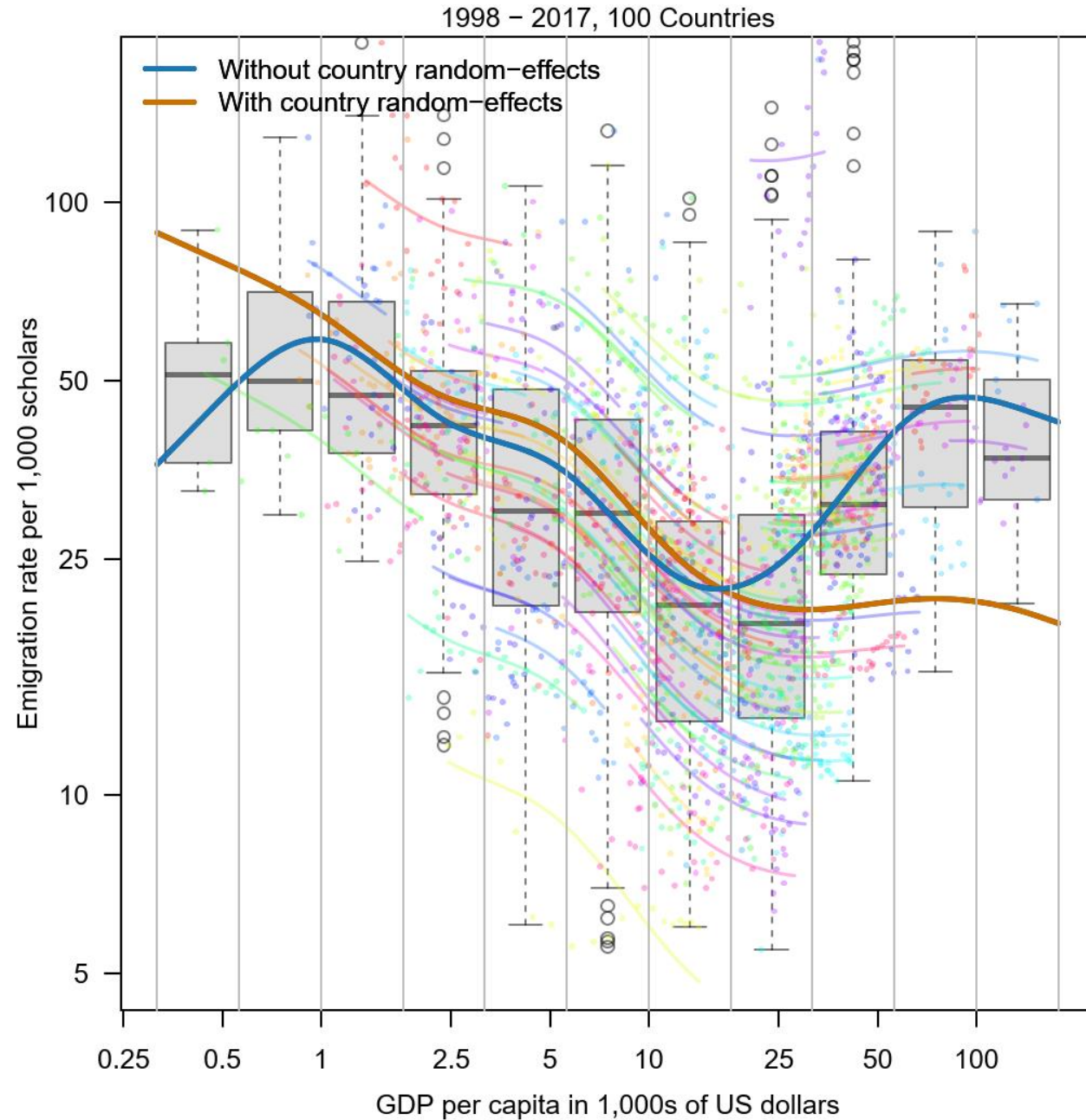


Table S2. List of countries used in the model sorted according to the average number of scholars. The first 20, 30, 40, ..., and 90 countries were selected for sensitivity analyzes.



Countries used in models

Nb.	Country	Scholars	Nb.	Country	Scholars
1	United States	669677.9	51	Colombia	4913.5
2	China	475115.1	52	Bulgaria	4172.8
3	Japan	206527.3	53	Indonesia	3805.0
4	United Kingdom	152962.8	54	Morocco	3798.3
5	Germany	122051.4	55	Slovenia	3187.6
6	Italy	109290.0	56	Bangladesh	3057.1
7	France	102041.2	57	Algeria	2991.7
8	India	100772.4	58	Vietnam	2301.3
9	Spain	89088.8	59	Lithuania	1983.0
10	South Korea	77643.4	60	Jordan	1814.8
11	Brazil	76850.8	61	Belarus	1796.5
12	Canada	73705.2	62	Kenya	1693.2
13	Russia	59155.2	63	Estonia	1447.6
14	Australia	59016.7	64	United Arab Emirates	1439.8
15	Netherlands	43707.4	65	Kazakhstan	1410.8
16	Turkey	37379.6	66	Lebanon	1396.9
17	Poland	32349.8	67	Peru	1337.4
18	Iran	28018.3	68	Philippines	1260.8
19	Sweden	27915.3	69	Uruguay	1121.7
20	Switzerland	26399.8	70	Ethiopia	1098.6
21	Belgium	21124.0	71	Iraq	1040.8
22	Mexico	20306.4	72	Kuwait	1033.6
23	Israel	17770.8	73	Nepal	1017.4
24	Denmark	16926.4	74	Iceland	961.5
25	Greece	16188.5	75	Latvia	930.9
26	Portugal	15873.5	76	Ghana	924.4
27	Czechia	15422.0	77	Cameroon	850.2
28	Austria	15296.1	78	Sri Lanka	846.0
29	Argentina	14918.5	79	Georgia	838.9
30	Finland	14385.2	80	Senegal	828.1
31	Malaysia	13327.5	81	Uganda	800.3
32	Norway	12781.5	82	Tanzania	775.8
33	Singapore	11946.5	83	Oman	748.4
34	Egypt	11802.1	84	Puerto Rico	741.5
35	Hong Kong SAR China	11524.0	85	Armenia	734.9
36	Hungary	10822.0	86	Qatar	727.1
37	New Zealand	10571.0	87	Cyprus	707.5
38	South Africa	10552.0	88	Bosnia & Herzegovina	652.7
39	Ukraine	10438.1	89	Uzbekistan	643.1
40	Pakistan	10117.5	90	Côte d'Ivoire	608.7
41	Ireland	9207.3	91	Ecuador	606.3
42	Romania	8968.1	92	Azerbaijan	591.5
43	Chile	8050.8	93	Luxembourg	525.0
44	Serbia	7010.5	94	Burkina Faso	503.4
45	Thailand	6600.2	95	Costa Rica	483.5
46	Croatia	6066.2	96	North Macedonia	473.7
47	Saudi Arabia	5626.6	97	Sudan	430.6
48	Tunisia	5330.3	98	Macao SAR China	429.5
49	Slovakia	5305.9	99	Zimbabwe	415.4
50	Nigeria	5091.4	100	Malawi	342.2

CONCLUSION

- A new database for the study of academic migration at the micro and macro levels
- Initial evidence that economic development does not increase the propensity of academic emigration in low- and middle-income countries
- A lot more to be considered: gender dimension, the role of institutions, the role of networks of collaborators, how to link this database to additional data sources, etc.

THANK YOU!

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